EXPERIMENT - 8

8. Perform basic Image Handling and processing operations on the image is to read an image in python and Dilate an Image using Dilate function

Aim

To read an image in Python and perform image dilation using the OpenCV dilate() function.

Procedure

1. Import the OpenCV and NumPy libraries.
2. Read the input image using cv2.imread().
3. Create a kernel using numpy.ones().
4. Apply dilation using the cv2.dilate() function.
5. Display the original and dilated images.
6. Observe the changes in the image.

Program

```
import cv2
import numpy as np

img = cv2.imread("input.jpg")

kernel = np.ones((5, 5), np.uint8)

dilated = cv2.dilate(img, kernel, iterations=1)

cv2.imshow("Original Image", img)
cv2.imshow("Dilated Image", dilated)

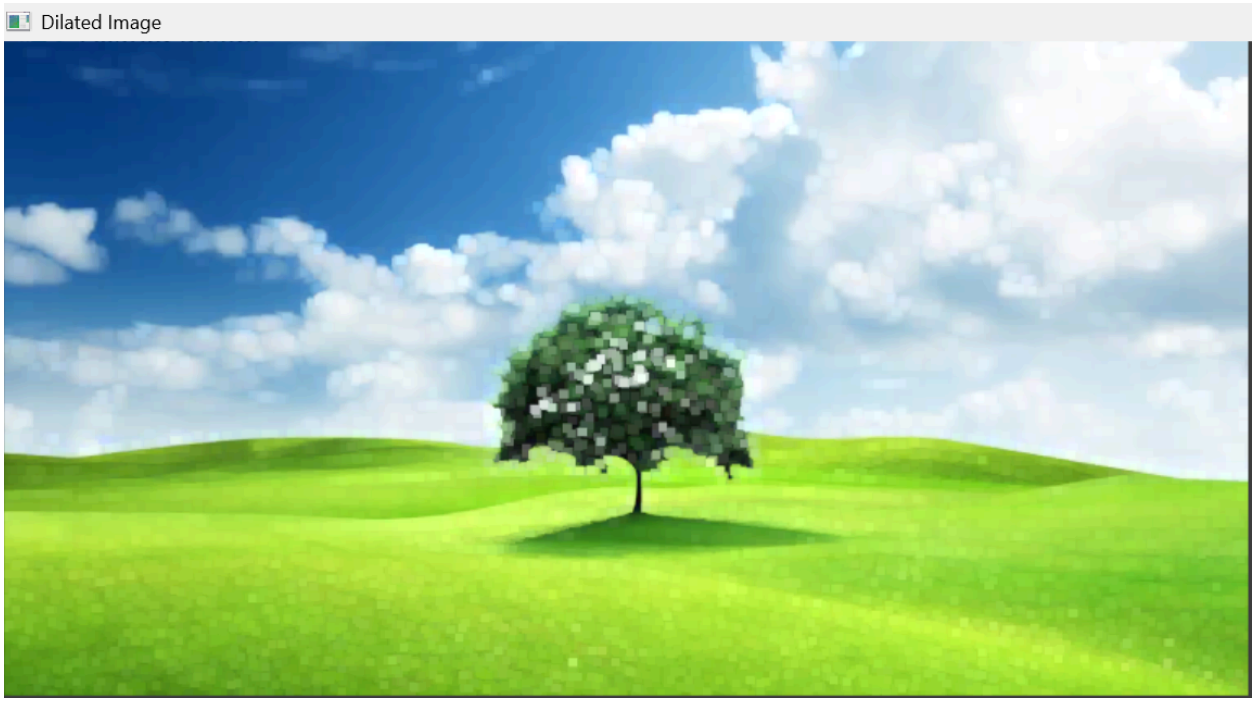
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Sample Input

Input.jpg



Sample Output



Result

Thus, the input image was successfully read and dilated using the OpenCV dilate() function.